Seojung **Min**

PH.D. CANDIDATE IN ROBOTICS (TACTILE PERCEPTION & DEXTEROUS MANIPULATION)

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Research Interests

I study **how robots interpret and use contact signals that arise when a grasped object (tool) interacts with the environment**. Using dual tactile sensors mounted on a gripper, I analyze **sliding contact, shear cues, and other extrinsic interactions** that reveal surface geometry or constraints through the tool. I build integrated robot–tactile systems (**UR5e + vision/magnetic tactile sensing**) to explore how such tool-mediated interactions can support robust, contact-rich manipulation.

Keywords Tool-mediated tactile perception · Extrinsic contact inference · Sliding/shear-based geometry & constraint estimation · Contact-rich robotic manipulation · Contact-driven perception

Education

KAIST(Korea Advanced Institute of Science and Technology)
Daejeon, S.Korea

PH.D. CANDIDATE IN MECHANICAL ENGINEERING *2022 - PRESENT*

- Advised by Prof. Jung Kim
- Thesis(Tentative): ***Touch-Based Extrinsic Contact Inference for Tool-Mediated Robotic Manipulation***
- Keywords: Vision-based tactile sensing, Extrinsic contact, Tool-tip control

M.S. IN MECHANICAL ENGINEERING *2013 - 2015*

- Advised by Prof. Jung Kim
- Thesis: ***Inertial Sensor-based Ambulatory Motion Sensing for Inverse Dynamics Analysis of Human Motion***
- Keywords: Wearable device integration, Rigid body inverse dynamics, Joint torque/force calculation

B.S. IN MECHANICAL ENGINEERING *2008 - 2013*

Skills

Robotics & Control	ROS, MoveIt, UR5e, Kinova, Robotiq gripper, low-level velocity control, MuJoCo, Unity
Tactile Sensing	Vision-based sensors (GelSight Mini, DIGIT, GelSlim 3.0), Magnetic sensors (eFlesh)
Programming & Software	Python (NumPy, OpenCV, Open3D, Pytorch, Tensorflow), Matlab/Simulink, LabVIEW, C/C++, Java
Hardware & Fabrication	PCB design (EasyEDA, KiCad), Microcontrollers (MyRIO, Teensy, Arduino, STM32), SolidWorks, Laser cutter
Other Tools	HoloLens2, OptiTrack, sEMG, IMU, Silicone fabrication

Selected Projects

Tool-Mediated Tactile Perception & Extrinsic Contact Inference (Ph.D. Research)
KAIST

2024 - PRESENT

- Developed full tactile exploration platform (UR5e + dual-GelSight Minis + Robotiq 2f-85) for tool-environment sliding, shear, indentation, and constrained motion.
- Designed experimental protocols and algorithm for localizing extrinsic contact point (tooltip position) through in-hand pose estimation.
- Collected large synchronized tactile-FT datasets, and developed a data-driven 3-D contact force inference model.

Tactile Sensor-Based Contact Property Estimation
KAIST, in collaboration with KIMM

LEAD RESEARCHER *2025 - PRESENT*

- Fabricated and calibrated open-source eFlesh magnetic tactile sensors.
- Collected tactile dataset labeled with roughness using DIGIT and Kinova robot arm, for roughness inference.

Robotic Grasp Perception Using Vision-Based Tactile Sensors
KAIST, in collaboration with ETRI

LEAD RESEARCHER *2022 - 2023*

- Fabricated and calibrated open-source GelSlim 3.0 tactile sensors.
- Built real-time pose/contact estimation pipeline, and conducted controlled experiments with OptiTrack ground-truth tracking.

Other Projects

Wearable Haptics & Proprioception Enhancement *KAIST, in collaboration with KIMM*

LEAD ENGINEER *2022 - PRESENT*

- Designed multi-channel vibrotactile feedback hardware (FPCB, IMU interface, Teensy controller).
- Developed motion-intent detection algorithms for proprioception augmentation.

Earlier Work (Human Motion, EMG, Exoskeleton, Sensors) *KAIST*

MEMBER *2013-2015*

- IMU-based inverse dynamics analysis for human motion
- Inertial sensor-based wrist-worn activity tracking system (KAIST & H3 System Co.Ltd.)
- Gesture recognition for sEMG interfaces (KAIST & Samsung Electronics)
- Bio-signal–based elbow force estimation models for exoskeleton applications (KAIST & LG Electronics)
- Muscle-supporting exoskeleton system with sEMG-based control

Publications

Journals

Yang, Sanghoon, Won Dong Kim, Hyunkyu Park, **Seojung Min**, Hyonyoung Han, and Jung Kim. ”In-hand object classification and pose estimation with sim-to-real tactile transfer for robotic manipulation.” IEEE Robotics and Automation Letters 9, no. 1 (2023): 659-666.

Conferences

Park, Sungbin, **Seojung Min**, Junhwan Choi, Min Jin Yang, Dayoung Yeom, Wonseok Shin, Chaeree Park, and Jung Kim. ”Preliminary Study of Portable Vibrotactile and Electrical Stimulation System for Knee Proprioception Enhancement.” In 2023 20th International Conference on Ubiquitous Robots (UR), pp. 800-804. IEEE, 2023.

Seojung Min, and Jung Kim. ”Inertial sensor based inverse dynamics analysis of human motions.” In 2015 IEEE International Conference on Advanced Intelligent Mechatronics (AIM), pp. 177-182. IEEE, 2015.

Na, Youngjin, Sangjoon J. Kim, **Seojung Min**, Changmok Choi, Sang Joon Kim, Sungho Jo, and Jung Kim. ”Personalized protocol to select usable movements for myoelectric pattern recognition.” In 6th European Conference of the International Federation for Medical and Biological Engineering: MBEC 2014, 7-11 September 2014, Dubrovnik, Croatia, pp. 29-33. Cham: Springer International Publishing, 2015.

Chang, Handdeut, Gwang Min Gu, **Seojung Min**, Hyosang Lee, and Jung Kim. ”Investigation of a tolerable time delay in SEMG-based elbow assistive device.” In 2014 Proceedings of the SICE Annual Conference (SICE), pp. 1382-1387. IEEE, 2014.

Korean Patents

Seojung Min, Sangjin Lee, Juhye Park, and Hyunsik Oh. ”ARCHITECTURE DESIGN OF BASE SYSTEM MODEL USING ENGAGMEMENT SIMULATION”, KR Patent Application No. 10-2018-0043505, filed Apr. 2018, granted Nov. 2018.

Seojung Min, Sangjin Lee, Juhye Park, and Hyunsik Oh. ”Method of managing and sharing object data based on spatial data for simulation of object-based engagements”, KR Patent Application No. 10-2018-0046248, filed Sep. 2018, granted Sep. 2018.

Professional Experience

Agency for Defense Development *Daejeon, S.Korea*

RESEARCHER, MODELING & SIMULATION TECHNOLOGY LAB. *2017 - 2019*

- Developed reusable and modular software components for naval weapon systems.
- Conducted system-level simulations to validate reliability and reusability of developed models.

SAMSUNG Electronics *Suwon, S.Korea*

ENGINEER, WASHING MACHINE EXPERT LAB. *2016 - 2017*

- Designed motor controllers and implemented wash algorithms.
- Optimized algorithm parameters based on extensive experimental testing.

Teaching

Teaching Assistant *KAIST*

COURSEWORK *2022 - 2025*

- My ME (ME493)
- Mechatronics System Design (ME203)
- Applied Electronics (ME207)
- Engineering Design (ME340)

Mentorship *KAIST*

UNDERGRADUATE RESEARCH PROGRAM *2022*

- *Development of surface electromyography interface and haptic device for metaverse applications*